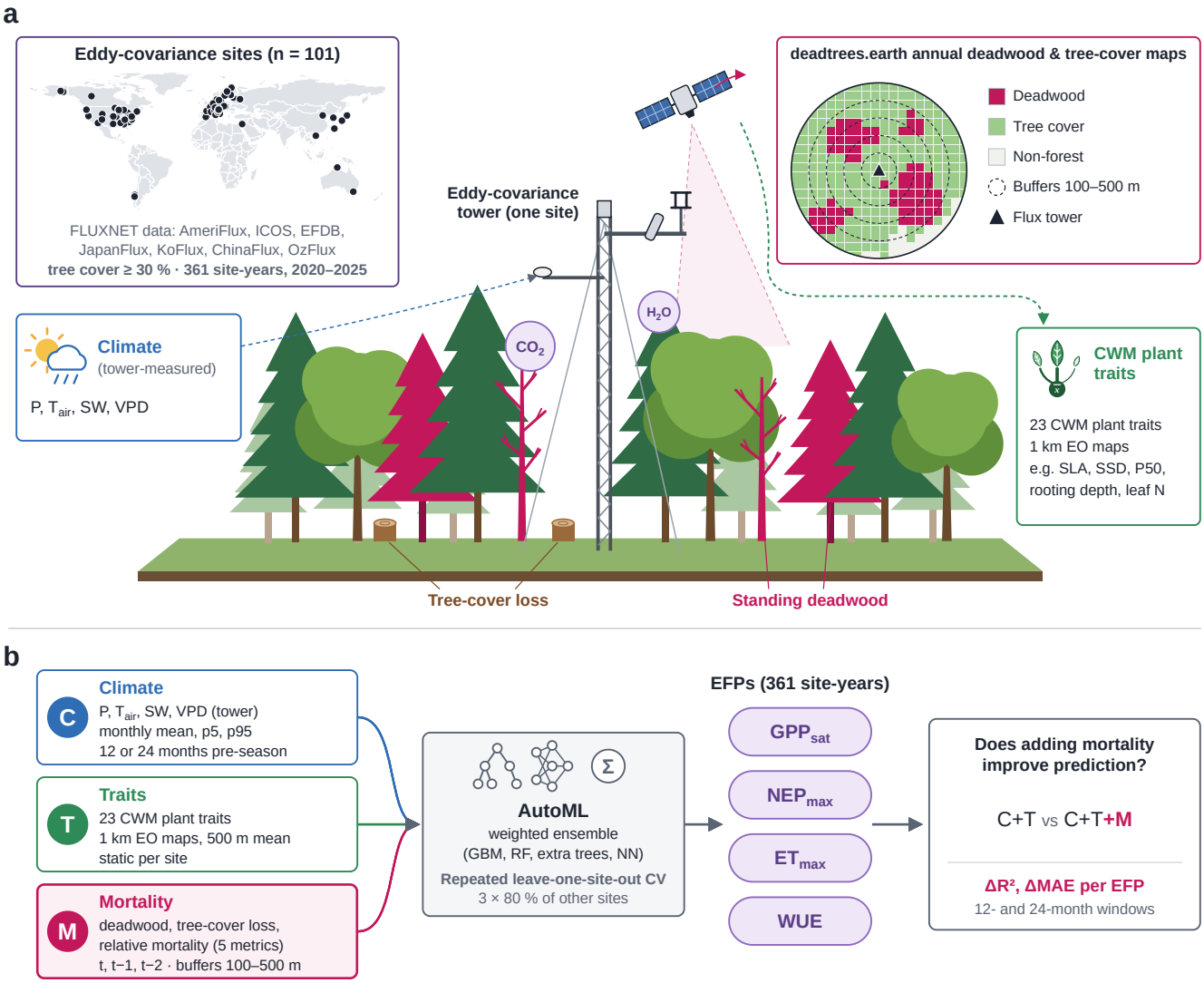

a
Eddy-covariance sites (n = 101)
FLUXNET data: AmeriFlux, ICOS, EFDB, JapanFlux, KoFlux, ChinaFlux, OzFlux
tree cover ≥ 30 % · 361 site-years, 2020–2025
deadtrees.earth annual deadwood & tree-cover maps
Deadwood
Tree cover
Non-forest
Buffers 100–500 m
Flux tower
Eddy-covariance tower (one site)
Climate
(tower-measured)
P, T air, SW, VPD
CO2
H2O
CWM plant traits
23 CWM plant traits
1 km EO maps
e.g. SLA, SSD, P50, rooting depth, leaf N
Tree-cover loss
Standing deadwood
b
C
Climate
P, T air, SW, VPD (tower)
monthly mean, p5, p95
12 or 24 months pre-season
T
Traits
23 CWM plant traits
1 km EO maps, 500 m mean
static per site
M
Mortality
deadwood, tree-cover loss,
relative mortality (5 metrics)
t, t−1, t−2 · buffers 100–500 m
AutoML
weighted ensemble
(GBM, RF, extra trees, NN)
Repeated leave-one-site-out CV
3 × 80 % of other sites
EFPs (361 site-years)
GPP sat
NEP max
ET max
WUE
Does adding mortality improve prediction?
C+T vs C+T+M
ΔR², ΔMAE per EFP
12- and 24-month windows